

Keeping Incarcerated People Safe? Evaluating PREA

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[GITHUB LINK]

1 Abstract

The 2003 Prison Rape Elimination Act (PREA) aimed to establish a zero-tolerance policy for rape in American prisons and jails. Little empirical work has yet directly examined PREA's impact. By looking at how states differed in compliance with the Act over time, we find that PREA increased allegations. In particular, an additional year of compliance is associated with 9 additional allegations per 1000 prisoners, from 2014-2020. We use publicly available Bureau of Justice Statistics data to measure the outcome, publicly available states governors letters to the DOJ to measure the treatment, and instrumental variables (IV) two-stage least-squares regression to attempt a causal estimate.

2 Introduction

Sexual assault in prisons is an unspeakable crime. When it happens due to the deliberate indifference of correctional authorities, it is an affront to fundamental values of fairness and justice, as highlighted by the Supreme Court Case *Farmer v. Brennan*. The Eighth Amendment of our Constitution formalizes this intuition, prohibiting cruel and unusual punishment, as interpreted by evolving standards of decency. Awareness from the public regarding the excesses of a punitive justice system, in addition to wider support for a more rehabilitative justice system following mass protests in 2020, reinforces societal opinion towards this often overlooked problem. As stated by nonprofit advocacy organization JDI International, which works to end all forms of sexual abuse within detention: "Rape is not part of the penalty." Yet America's mass incarceration problem persists — and prisons that are unable to be humane in providing shelter or adequate medical care, much less a continuum of reentry programming that supports the reintegration of inmates into the community, are also likely to turn a blind eye to sexual abuse perpetuated within

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confinement. Congress wrote in 2003 that "experts have conservatively estimated that at least 13 percent of the inmates in the United States have been sexually assaulted in prison," and that the total number of inmates sexually assaulted in the last 20 years was likely over 1 million (34 USC 30301). They stated that younger incarcerated people were more at risk of being assaulted, that sexual assault in prison inhibits successful reentry, escalates interracial tensions, and that "members of the public and government officials are largely unaware of the epidemic character of prison rape and the day-to-day horror experienced by victimized inmates." (34 USC 30301). The need for comprehensive federal oversight and guidance on the problem resulted in the passage of the Prison Rape Elimination Act (PREA) in 2003.

"The Prison Rape Elimination Act (PREA) is perhaps the most significant law reform project undertaken on U.S. prison issues in the twenty-first century," writes lawyer and advocate Gabriel Arkles in a critical 2015 article. PREA establishes standards for confinement aimed at eliminating the problem of sexual assault within American prisons, as well as establishing mandatory audits and enforcement mechanisms to ensure state and local compliance.

PREA continues to have wide-ranging effects on the welfare of Americans in confinement. In particular, PREA establishes yearly audit requirements for state prisons and jails, with separate requirements for adults and juveniles, staffing, training, reporting accessibility, and regulations involving security cameras. Importantly, PREA requires that correctional authorities maintain "separation between inmates at risk of being sexually victimized and inmates at risk of being sexually abusive" while making housing or programming decisions, such as deciding who bunks with who (PREA Standard 115.42). In particular, at minimum, PREA requires correctional authorities in prisons and jails to consider ten different criteria for identifying potential victims of abuse and separating them from potential abusers, including: disability status, age, physical build, previous incarceration status, nonviolent previous criminal history, prior convictions for sex offenses, LGBT status or perceived LGBT status by others, previous victimization experience, inmates' own perception of vulnerability, and inmate immigration status (PREA 115.41). Then, in a mandatory reassessment, not to exceed 30 days from the inmate's arrival at the facility, facilities reassess the inmate's risk of victimization or abusiveness based upon any additional, relevant information received by the facility since the intake screening. Similar requirements apply to community confinement (halfway houses), juvenile facilities, and to some extent lockups, local jails for temporary detention. As part of comprehensive standards across prevention planning, training, education, reporting, data collection, and more, PREA limits cross-gender viewing and searches, designates an upper-level official of any correctional agency to serve as a coordinator, and protections from retaliation for staff and inmates who report abuse.

Unfortunately, funding to the National PREA Resource Center was cut in April 2025 as part of sweeping DOJ grant cancellations by the Trump Administration (Brennan Center for Justice). The PREA Resource Center also oversaw the PREA audits, the nation's only oversight tool monitoring compliance with federal rules to end sexual abuse behind bars (Just Detention International 2025).

Although BJS publishes state-level data on sexual abuse within prisons and jails, we are not aware of any independent empirical analyses on the direct impact of the passage of PREA on reducing the prevalence of sexual assault within American prisons and jails. As funding for PREA resources are threatened by the Trump Administration, we believe an empirical demonstration of the lifesaving effects of PREA is necessary, if possible.

In addition to empirical analysis, the paper will illustrate why justice reform programs like those funded by PREA, which are intended to respect the dignity and safety of incarcerated people, are important and often lifesaving. The data demonstrate how PREA and associated programs help keep incarcerated adults and children, who are disproportionately racial minorities, safe and maintain oversight and transparency within correctional facilities.

Preliminary analysis shows that in many states, reported rates of both allegations and substantiated incidents ¹ have increased since the passage of PREA, with a sharp increase at the inclusion of new federal standards. This does not mean that PREA did not work. Since PREA both establishes a standard for improved reporting and data collection and attempts to prevent prison rape itself, the increase in reported incidents after the passage of the act and after release of updated standards in 2012 does not contradict the possibility of real-world improvement after PREA. After all, experts agree that incidence rates were likely underreported due to fear of retaliation before PREA. However, estimating the true benefits of PREA are difficult for this reason.

To solve this problem, we apply a method of data analysis known as multivariate regression, which is commonly used in econometrics to estimate the impacts of policies. Without being able to randomly assign PREA to prisons, we can still investigate the real impact of PREA by comparing different groups of prisons (states) that experienced different levels of exposure to the policy, or the *treatment*. For example, if allegations increased over time for Delaware but not for Massachusetts, but Delaware has many more years of reported compliance with PREA than Massachusetts, we can conclude that continued compliance with the policy has a positive influence on the number of allegations in a given state. This is implemented by regressing $Y_{s,t}$, the outcome in state s and at period t , on state fixed effects, period fixed effects, and $D_{g,t}$, the treatment of group g at period t , which we define as cumulative years of state compliance with PREA. When applied to BJS data from 2012 to 2020, controlling for population, incarceration rate, violent crime rate, staff per 1000 inmates, and year fixed effects, this method of data analysis shows that PREA indeed influenced counts of allegations, and can produce a numerical estimate of the size of the effect. Please refer to Section 4 for a detailed description of the analysis, including its assumptions and caveats.

Section 3 will outline the policy context of PREA since its passage 23 years ago. Section 4 will explain the statistical methods used to estimate the impact of PREA. Results, robustness checks and limitations of our methods are discussed in Section 5. Finally, policy implications and directions for further research

¹BJS incident rates per thousand prisoners.

are laid out in Section 6.

3 Background

The grassroots movement that brought about the Act in 2003 aimed to establish a zero-tolerance policy for sexual abuse within American prisons and jails.

Literature has arisen to study the impacts of the Act as well as the causes and magnitude of the phenomenon on the whole yet few empirical studies on the efficacy of the Prison Rape Elimination Act exist. Practitioners, social workers, and academics have conducted cross-disciplinary studies of the implementation of PREA training and audits across the nation. Smith et al. 2025 analyzes inmate and staff perceptions towards PREA in an effort to identify why incident counts have risen since PREA was passed; they find demographic variation in support for PREA: inmates are less likely to support PREA than officers; transgender and longer sentenced-inmates are less likely to support PREA; relatively high support and knowledge for PREA in both inmates and correctional officers; higher support for PREA reported among female, African-American, and liberal-minded correctional officers (Smith et al. 2025). A 2024 mixed methods survey of three prisons in the Southeast US found Black inmates statistically less likely to report, more likely to fear retaliation, less positive that PREA has an effect, and inversely, that Black staff are statistically more likely to report (Ferdik, Smith 2024).

It is important to note that the Act is not legally binding on states, influencing its effectiveness. "In most cases where prisoners raise violations of PREA in their complaints, courts decline to consider PREA at all because of the lack of a private right of action" (Arkles). Similarly, Giovanna Shay warned in the Northeastern University Law Journal that "the threatened loss of funding is simply too weak a compliance mechanism to create much incentive for some governors to implement the regulations," echoing sentiments of legal experts and policymakers at the time (Shay 2014). In Thompson 2021, a legal argument is made on behalf of compliance with PREA, where the author notes that Arkansas was one of two states at the time that had neither submitted an assurance or a certification of compliance with PREA — ever! After all, as reported by a spokesperson for the governor, who refused federal PREA audits, the penalty for failure to comply would be less than what was required to come into compliance with PREA: less than \$100,000² (Thompson 2021). This fact further motivates the need for an empirical analysis of whether or not PREA has had its intended impact on prisons and jails.

Is this analysis even possible? After all, adverse health outcomes behind bars are underreported [citation needed]. In a study facing similar challenges, Alsan and Yang found that randomization of healthcare accreditation for prisons and jails "reduces mortality among the incarcerated," a mortality rate "which is three times higher among control facilities than official estimates suggest." (Alsan and Yang 2025). Second, the study of an outcome like prison rape is inherently difficult due to validity concerns about the data, especially where inmates fear retaliation from authorities or other inmates (Dumond). Social pressure

²As of 2022 and 2023, Arkansas has submitted assurances towards certification.

against reporting inmate-on-inmate harassment, including inmate-on-inmate sexual abuse, often involves threats of violence or actual violence, up to and including murder. It may be worth wondering if official data is a reliable source of information regarding sexual abuse in confinement at all.

However, as Massachusetts Department of Corrections counselor Bob Dumond testified: "If they feel that the institution will respond appropriately, inmates will come forward. They want this to stop. They don't want to be victimized anymore," (Dumond, date unknown). In the data, we do see a significant aggregate increase in both allegations and substantiated incidents since PREA was passed (Figures 1 and 2). Although we don't know *a priori* if this increase can be attributed to the passage of PREA, we do know that PREA has the potential to improve reporting of incidents by inmates, despite the fear of retaliation by others. Thus, although a causal estimate of PREA's effects from BJS data may be imprecise, it is far from useless.

3.1 Preliminary Trend Analysis

1. Both allegations and substantiated incident counts have risen since the passage of PREA, with a sharp increase at the inclusion of new federal standards. Since PREA both establishes a standard for improved reporting and data collection, as well as has the potential to decrease incident counts itself, the increase in reported incidents after the passage of the act and after release of updated standards in 2012 does not contradict the possibility of real-world improvement after PREA. The sharp increase in 2013, with a plateau in 2015, is consistent between alleged and substantiated case counts (Figures 1,2).
2. States varied considerably in the amount of incidents they reported, either alleged or substantiated. ([State Heatmap](#)).

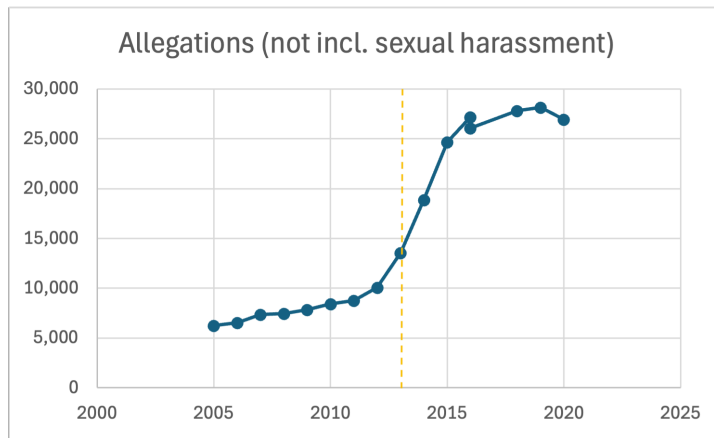
4 Data and Empirical Strategy

Since 2003, certain American states have agreed to fulfill the requirements of the PREA legislation to secure a fraction of certain funds allocatable through the Department of Justice. Others have declined, or agreed in one year but declined in successive ones.

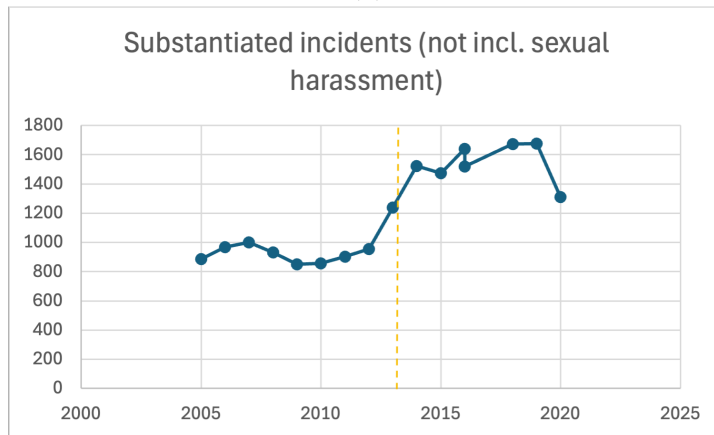
States have three options:

1. Comply with standards.
2. Promise 5% of DOJ-allocated funds towards compliance.
3. Decline altogether and lose 5% of funds.

We measure state variation in exposure to the policy using the cumulative count of years that a state complies with PREA standards in the study period. In other words, the number of years from 2012 to 2020 that a state complies with option (1). This program rollout design, which evaluates the effects of a



(a)



(b)

Figure 1: Annual alleged and substantiated counts of sexual victimization reported by Federal and State authorities (BJS 2020)

program based off of participants' varying exposure to the program, has been used by economists to evaluate the Food Stamp Program (Hoynes and Schanzenbach 2016), Medicare (Almond, Chay, and Greenstone 2007; Finkelstein and McKnight 2008), Women, Infants, and Children (WIC) ((Hoynes, Page, and Stevens 2011), family planning programs (Bailey 2012), Title I (Cascio et al. 2010), and community health centers (Bailey and Goodman-Bacon 2015).

Updated PREA standards were released in 2012, and states are given two years to comply with them. Accordingly, starting 2014, the Office of Justice Programs Assistant Attorney General sent letters every March to states, territories, and the District of Columbia, on behalf of the DOJ, explaining states' responsibilities to PREA. Every state responds with one of the three options outlined above, and this information is available through the Bureau of Justice Assistance (BJA) website, beginning in 2012, which is when the final rule for PREA was published.

We identify only reported compliance with standards as our treatment; we treat a promise to comply (an assurance) identically to declining to submit an assurance or certification, because the parameter we are interested in is the effect of being in compliance with PREA. According to the BJA letters, from 2014-2024, Alaska, American Samoa, Arkansas, California, Colorado, Guam, Idaho, Indiana, Michigan, Puerto Rico, South Carolina, U.S. Virgin Islands, and Wisconsin are never-compliers, whereas Missouri, New Hampshire, New Jersey, and North Dakota are top compliers with 10 years of compliance each.

The primary outcome to be measured is the number of allegations and the number of substantiated

incidents of sexual assault per 1,000 prisoners. In the panel data, the Bureau of Justice Statistics classifies both alleged and substantiated sexual abuse incidents into four types: Inmate-on-inmate nonconsensual sexual acts, inmate-on-inmate abusive sexual contact, staff-on-inmate sexual misconduct, staff-on-inmate sexual harassment. For the purposes of this analysis, we sum the counts of these incidents to produce state counts of incidents per 1000 prisoners.

Linear regressions, commonly used in the social sciences, estimate the relationship between a single value, known as the dependent variable, and one or more explanatory variables. They can produce coefficients describing the strength of the relationship between one of the explanatory variables and the dependent variable. By combining our knowledge of official BJS numbers on prison sexual abuse, as they fluctuate year-by-year, with a state's level of compliance with the Act, along with other statistical factors that may influence the prevalence of the phenomenon, we can estimate a single number that describes the effect of a state adopting full compliance with the Act on the incidence of sexual assault in prisons. In particular, in addition to indicators of incarceration rate, violent crime, state population, and correctional officer employment, we include terms that capture unobservable characteristics common to each state across time, and common to each year across states: this technique is known as a two-way fixed effects regression. We can identify how good our estimate is, in an abstract sense, using the p-value, which identifies the probability that the relationship between the outcome variable and the dependent variable occurred by pure chance.

Thus, the slope of our regression will tell us how much compliance affects the level of sexual abuse outcomes in prisons, which is exactly what we want. However, in our case, a simple linear regression (OLS) is misleading. A simple linear regression is misleading because any given state will not be randomly assigned years of compliance. Instead, states may comply for more years because they want to fix very poor outcomes within their own system, or they may comply for more years because they know they already have superior outcomes within their system. States may comply for less years because they want to avoid reporting the very poor outcomes within their own system, or they may comply for less years because they are unable to comply because their prison conditions, organization, staffing culture, and so on, are already terrible. Therefore, to accurately estimate the effect of compliance on outcomes, we need some outside way of predicting how many years a given state will comply.

We turn to an econometric approach known as an instrumental variables (IV) design. For example, in a 1991 study, economists Angrist and Krueger exploit randomness in student birthdays to estimate the impact of compulsory schooling on future wages, since one's birthday does not affect one's wages but it affects whether or not one gets an additional year of schooling because mandatory schooling ends when one is past a certain age, such as sixteen (Angrist and Krueger 1991).

- First, they show that people born in the beginning of the year will have less schooling, on average, than those born near the end of the year.
- Then, they look up the weekly earnings of people born early and compare them to the earnings of people born late. People born late happen to earn slightly higher.

- They divide that difference by the ratio by which they differed in schooling. This tells them how much schooling affects earnings later in life.

Therefore, if we can identify an outside factor that affects the level of the outcome variable through only years of compliance, and that doesn't affect the outcome variable otherwise, we can see first how much the factor affects years of compliance per state, and second how much years of compliance per state influences alleged and substantiated incident counts per state. Finally, we can divide the outcome for each state by the level of this outside factor to identify just the effect of additional years of compliance on each state. We choose whether or not a state has ever been compliant with PREA in a given year, a binary variable, as this outside factor: our instrument.

- First, we show that state-years that have ever been compliant will have more cumulative compliance years, on average, than state-years that are not.
- Then, we look up the rates of sexual abuse within that state for that year, as compared to state-years that have not been compliant.
- We divide this difference by the ratio by which that state year differed in cumulative compliance years.

For each state and year, e.g. {(Arizona, 2012) (California, 2014), (California, 2018)}, we regress current treatment status on ever-treated status:

$$CurrentlyTreated_i = \phi EverYetTreated_i + \mu_i + \delta_t + \epsilon_t$$

ϕ will tell us the probability of being currently treated having ever been treated, or formally, $P_i(\text{currently treated} \mid \text{evertreated})^3$. In turn, this can tell us what the effect of starting to comply is on a state ever complying. This will provide context for the second-stage regression, which asks in a given year what is the effect of additional years of compliance on the level of substantiated or alleged incidents within a state's prisons and jails. Formally, the empirical strategy is an instrumental variables differences-in-differences (DiD) design. As a note, many instrumental variables designs, like Angrist and Krueger's 1991 study, use a binary treatment variable on a cross-sectional study design, whereas our treatment variable, cumulative compliance years, is continuous. This choice of treatment is justified: we are interested in investigating, what the effects of a given State's continued compliance with PREA, which is legally-binding, may be, at the decision point of a state governor deciding whether or not to come into compliance with PREA's standards. Because our treatment (cumulative years of compliance) is not binary but varies in intensity, following Angrists and Imbens 1995, we identify an Average Causal Response (ACR) identified as "a weighted average of per-unit causal effects along the length of an appropriately defined causal response function" (Angrist and Imbens 1995).

Although the two-way fixed effects regression methodology, used within our instrumental variables design, is applied across the social sciences to study crime, labor, and health topics, recent literature has

³In other words, the first stage estimate.

questioned the validity of causal estimates obtained as such. In particular when the size of the effect of a policy varies across groups, caution should be used interpreting a two-way fixed effects regression coefficient on the treatment variable as the average treatment effect (ATE) (de Chaisemartin and D’Haultfœuille 2020). When group and time fixed effects are present, this number is shown to be the ”weighted average of all possible two-group/two-period DD estimators in the data.” (Goodman-Bacon 2021). However, ”when treatment effects do not change over time, TWFEDD yields a variance-weighted average of cross-group treatment effects and all weights are positive.” (Goodman-Bacon 2021). In this case, we are specifically interested on the impact of *continued* compliance with PREA on substantiated incident rates by state: the cumulative years of compliance variable is continuous. This allays the concern regarding negative weights and average treatment effect.

What kinds of impacts should we expect from PREA, in terms of direction and magnitude? Again, our aim is to investigate using causal inference the nature of the impact that PREA has had on the American prison system. On one hand, consistent with a ”reporting hypothesis,” states with more compliance-years may see more counts of incidents than states with less compliance-years, after controls. On the other hand, the ”law-and-order hypothesis” is that states with more compliance-years will see less incidents than states with less-compliance years. In Section 5, we present results that are mostly consistent with the former hypothesis. However, we find that our sample is insufficiently powered to investigate the impact of PREA on substantiated incidents, which are more rare. This highlights the need for further research as additional data becomes available.

Detailed data sources are outlined in Figure 4.

Table 1: Descriptive Statistics: State-Level Average Rates (2014–2020) and Pooled Compliance Years

Variable	Mean	Median	S.D.	Min.	Max.	<i>N</i>
Alleged per 1,000 (state avg.)	16.651	14.410	11.387	4.000	64.268	46
Substantiated per 1,000 (state avg.)	1.509	1.037	1.422	0.086	6.160	46
Cumulative PREA compliance years (pooled)	0.939	0.000	1.479	0.000	7.000	296

Notes: Alleged and substantiated rows summarize annual per-1,000 rates over 2014–2020 (across all states). The compliance row pools all state-year observations with non-missing cumulative compliance years in 2014–2020; *N* is the number of state-year observations.

In the second stage, the following panel regression models are estimated using state-year observations:

4.1 Alleged Sexual Victimization Rate

$$\begin{aligned}
z_{st} = & \beta_0 \hat{\theta}_{st} + \beta_1 X_{1s} \\
& + \beta_2 X_{2st} + \beta_3 X_{3st} + \beta_4 X_{4st} \\
& + \phi_s + \gamma_t + \varepsilon_{st}
\end{aligned} \tag{1}$$

where z_{st} is the number of alleged incidents of sexual victimization per 1,000 prisoners in state s and year $t - 1$, X_1 is population, X_2 is violent crime rate, X_3 is staff per 1000 inmates, X_4 is state incarcerated per 100,000 population, and ϕ and γ are time and group (year and state) fixed effects.

This is the R command for regression (1):

```
iv_model_alleged_lag <- lm(
  alleged_per_1000_tminus1 ~ cumulative_years_hat + population_millions +
  incarceration_rate_per_100k + staff_per_1000_inmates +
  violent_crime_rate_per_100k + factor(state) + factor(year),
  data = iv_data_lag
)
```

4.2 Substantiated Sexual Victimization Rate

$$\begin{aligned}
 y_{st} = & \beta_0 \hat{\theta}_{st} + \beta_1 X_{1s} \\
 & + \beta_2 X_{2st} + \beta_3 X_{3st} + \beta_4 X_{4st} \\
 & + \phi_s + \gamma_t + \varepsilon_{st}
 \end{aligned} \tag{2}$$

where y_{st} is the number of substantiated incidents of sexual victimization per 1,000 prisoners in state s and year $t - 1$, X_1 is population, X_2 is violent crime rate, X_3 is staff per 1000 inmates, X_4 is state incarcerated per 100,000 population, and ϕ and γ are time and group (year and state) fixed effects.

4.3 Variable Definitions

Treatment Variable:

- $\theta_{st} : \sum_{t=2013}^{2020} \mathbf{1}\{\text{State } s \text{ certifies to DOJ in period } t\}$

This will be referred to as *cumulative compliance years*. As implemented in the second-stage regression, these are not the actual counts of cumulative years, but the fitted cumulative compliance years estimated in the first stage regression, and are referred to as $\hat{\theta}$.

Confounders:

- Population_s: State population in millions (2020 Census).
- Incarceration Rate_{st}: Number of prisoners per 100,000 state population in state s and year t .
- Violent Crime Rate_{st}: Violent crime rate per 100,000 population in state s and year t .
- Staff per 1,000 Inmates_{st}: Number of correctional staff per 1,000 prisoners in state s and year t .
- γ_t : Year fixed effects for years 2012-2019 (base year: 2012).
- ε_{st} : Error term.

4.4 Data

Two key annual datasets are used. The first is the panel dataset for the primary outcome and are the BJS Surveys on Sex Victimization. These data are publicly-available reports from 2012 to present, and include counts by correctional staff and PREA auditors on the number of incidents per facility and organized by

state. The second dataset, for the key independent variable, was compiled out of yearly lists of PREA certifying states on the BJA website.

Experts agree that prison rape is likely underreported, which makes the question of false negatives key. As the BJS reports in 2007, “inmate self-reports provide a basis for comparing and ranking facilities.” For the purpose of our analysis, although pre-2007 surveys of administrative records could not provide reliable estimates of sexual violence due to being limited to incidents reported to correctional authorities, starting from the 2007 NIS, conducted by RTI International, provided anonymity through a touch-screen, headphones-based interview format that allowed participants to complete the survey themselves. However, these estimates are not comparable to data past 2012, because they measure different outcomes. The beginning of the PREA audit cycles in 2012, as shown in the BJA letters, coincides with increased visibility and accessibility of reporting methods within prisons and jails due to PREA-related changes. Thus, our hope is that the 2012-present BJS data is accurate for the model at hand. We report a lagged outcome, meaning the outcome variable for year $(t-1)$ is regressed on the treatment variable for year t . This is because governors letters of certification refer to the prior audit year: for example, the FY 2016 letter, due in March of 2016, asked about compliance status as of August 2015. These letters begin in 2014, while 2020 is the last year for which a public Survey on Sexual Victimization is available from BJA. This restricts our sample window to 2014-2020.

As discussed above, for our design to be valid, ever-being-treated must be exogenous from levels of sexual abuse within a state, except by influencing additional years of compliance. We address this issue further by controlling for state-specific covariates. As discussed above, these covariates allow us to isolate the effect of PREA compliance, essentially cancelling out the effects of variation in other relevant variables as it relates to our parameter of interest. These covariates were determined *a priori* and their sources are described in Figure 4. Our Staff per 1,000 inmates is calculated from correctional officer employment, from the Occupational Employment and Wage Statistics (OEWS) and using corresponding state prisoner population counts. Violent Crime Rate is calculated from the FBI Uniform Crime Reporting program annual tables. Incarceration Rate is calculated directly from the BJS Surveys of Sexual Victimization that are used for our outcome variable.

Additionally, fixed effects are included in the specification for both state and year units. In other words, for each factor level (i.e. each subject), a subject-specific mean of our dependent variable (substantiated incident count) will be estimated. This is a fixed-effect (FE) coefficient and accounts for unobserved characteristics. Year FE are included to control for time-varying factors common across all states. A two-way FE regression therefore controls for both unobserved time-varying and state-varying factors. As a note, standard errors are clustered at the state level following best practices in inference (Bertrand, Duflo, and Mullainathan 2004).

5 Results

Table 2: First Stage: Cumulative PREA Compliance Years (Lag-IV Sample)

	Coefficient	Std. error	p-value
Ever treated with PREA compliance (through year t)	0.789***	(0.257)	0.002
Population (millions)	-0.041	(0.107)	0.703
Incarceration rate (per 100,000)	-0.001	(0.007)	0.897
Staff per 1,000 inmates	0.002	(0.003)	0.415
Violent crime rate (per 100,000)	0.000	(0.002)	0.965
Observations	236		
Dependent variable	Cumulative PREA compliance years		
State fixed effects	Yes		
Year fixed effects	Yes		
Clustering	State		
R ²	0.852		
Adjusted R ²	0.811		

Notes: The sample and covariates match the lag-outcome IV second stage for alleged incidents per 1,000 (outcome in $t - 1$). Approximate clustered Wald F -statistic for **EverTreated** (state-clustered HC1) is 9.434. Significance stars apply to the coefficient column: $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Table 3: IV/2SLS Estimates: Alleged Inmate-on-Inmate Sexual Victimization per 1,000 Prisoners (Outcome in $t - 1$)

	Coefficient	Std. error	p-value
Instrumented cumulative PREA compliance years	8.882**	(4.303)	0.040
Population (millions)	1.494	(1.172)	0.204
Incarceration rate (per 100,000)	0.034	(0.068)	0.617
Staff per 1,000 inmates	-0.013	(0.029)	0.657
Violent crime rate (per 100,000)	-0.023	(0.037)	0.533
Mean of dependent variable	16.004		
Observations	236		
State fixed effects	Yes		
Year fixed effects	Yes		
Clustering	State		

Notes: Dependent variable is alleged incidents per 1,000 prisoners in year $t - 1$. Treatment is cumulative years of PREA compliance instrumented by an indicator for ever having been compliant through year t (**EverTreated**). Covariates are listed in the table. Significance stars apply to the coefficient column: $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Table 4: IV/2SLS Estimates: Substantiated Inmate-on-Inmate Sexual Victimization per 1,000 Prisoners (Outcome in $t - 1$)

	Coefficient	Std. error	p -value
Instrumented cumulative PREA compliance years	0.237	(0.532)	0.656
Population (millions)	-0.668***	(0.132)	0.000
Incarceration rate (per 100,000)	0.005	(0.008)	0.522
Staff per 1,000 inmates	-0.001	(0.003)	0.844
Violent crime rate (per 100,000)	-0.008*	(0.004)	0.069
Mean of dependent variable		1.532	
Observations		236	
State fixed effects		Yes	
Year fixed effects		Yes	
Clustering		State	

Notes: Dependent variable is substantiated incidents per 1,000 prisoners in year $t - 1$. Treatment is cumulative years of PREA compliance instrumented by an indicator for ever having been compliant through year t (**EverTreated**). Covariates are listed in the table. Significance stars apply to the coefficient column: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

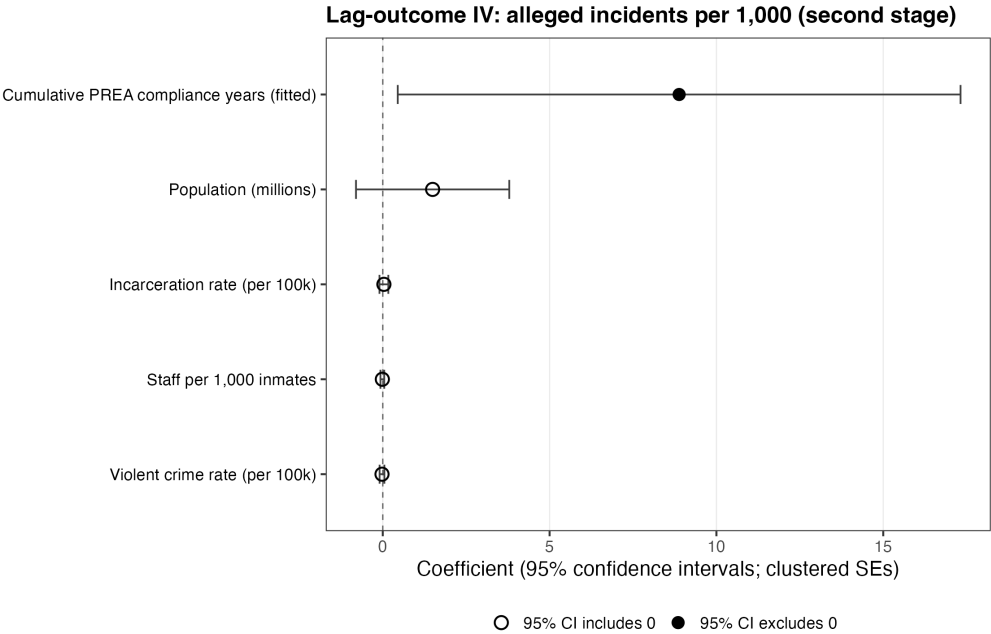


Figure 2: Second-stage regression coefficients for allegations per 1000 prisoners on instrumented cumulative compliance years (2014-2020).

5.1 Robustness Checks

Different researchers may look at the publicly available Bureau of Justice Statistics data and disagree on how effective PREA is. To help address this, I will provide several alternative models to show how different calculations of PREA’s effectiveness might produce different results.

- Standard Ordinary Least Squares (OLS) regression with fixed effects:

Alleged ($p = 0.48$):

$$\beta_{0st} = -0.7 \pm 1.0$$

Substantiated ($p = 0.25$) :

$$\beta_{0st} = -0.1 \pm 0.1$$

In the standard OLS regression specification, we cannot reject the null hypothesis.

- IV 2SLS regression without lag:

Alleged ($p = 0.21$):

$$\beta_{0st} = -4.0 \pm 3.1$$

Substantiated ($p = 0.94$):

$$\beta_{0st} = -0.0 \pm 0.3$$

In the IV 2SLS regression without lag, we cannot reject the null hypothesis.

Therefore, the result that PREA causes more allegations must be interpreted with caution because they are not consistent across alternative specifications.

5.2 Limitations

A major limitation of our study was that it was insufficiently powered to study the rate of substantiated incidents (Table 3). Given our identification strategy, the sample of state-years simply was not large enough to identify the effect of PREA on substantiated incidents. Our strategy restricts the study window to 2014-2020. However, an identical study design may be implemented upon the release of annual counts of victimization from 2020-2024, more recent Survey of Sexual Victimization, as governors’ certification status are available through 2024. This would represent a significant increase in the sample size.

6 Conclusion

Consistent with an increased reporting effect, we find that full compliance with Prison Rape Elimination Act standards is associated with a statistically significant increase of, on average, 9 allegations per 1000 prisoners in American federal and state prisons.

Although this result should be interpreted with caution as it is not robust across alternative specification, the effect size is strong. Around 16 allegations per 1000 prisoners is the mean reported across all

states for the study window; an increase of 9 allegations represents a substantial increase in reports of sexual abuse. Furthermore, the model itself is significant at a standard significance level of $p < 0.05$ (Figure 5).

We fill a gap in the literature by attempting a causal estimate of PREA and its intended effects. This information is relevant to states governors' offices, policymakers and academics in the criminal-legal space, as well as advocates and criminal-legal system involved individuals. The increase in allegations linked to additional compliance with the Act raises a serious concern: that sexual abuse within confinement continues to be vastly underreported. This point is reiterated in the qualitative literature by correctional researchers, psychologists, concerned segments of the law-enforcement population, and more.

We conclude with a note on the current state of compliance with PREA. In 2014, only two states – New Hampshire and New Jersey – certified compliance with PREA. In 2015, 11 did. In 2023, and 2024, 27 states did. Full compliance with PREA and federal funding for PREA audits, trainings, and programs is lifesaving and the results of this analysis demonstrate the enduring impact of PREA as well as the substantial room for improvement regarding this issue. As public beliefs shift regarding sexual abuse within confinement, and correctional cultures shift towards becoming less punitive, additional robust causal research on this question is urgent. Additional rigorous, transparent empirical research surrounding "what works" in criminal justice, using the highest quality evidence possible, and best practices in causal inference, is more necessary than ever.

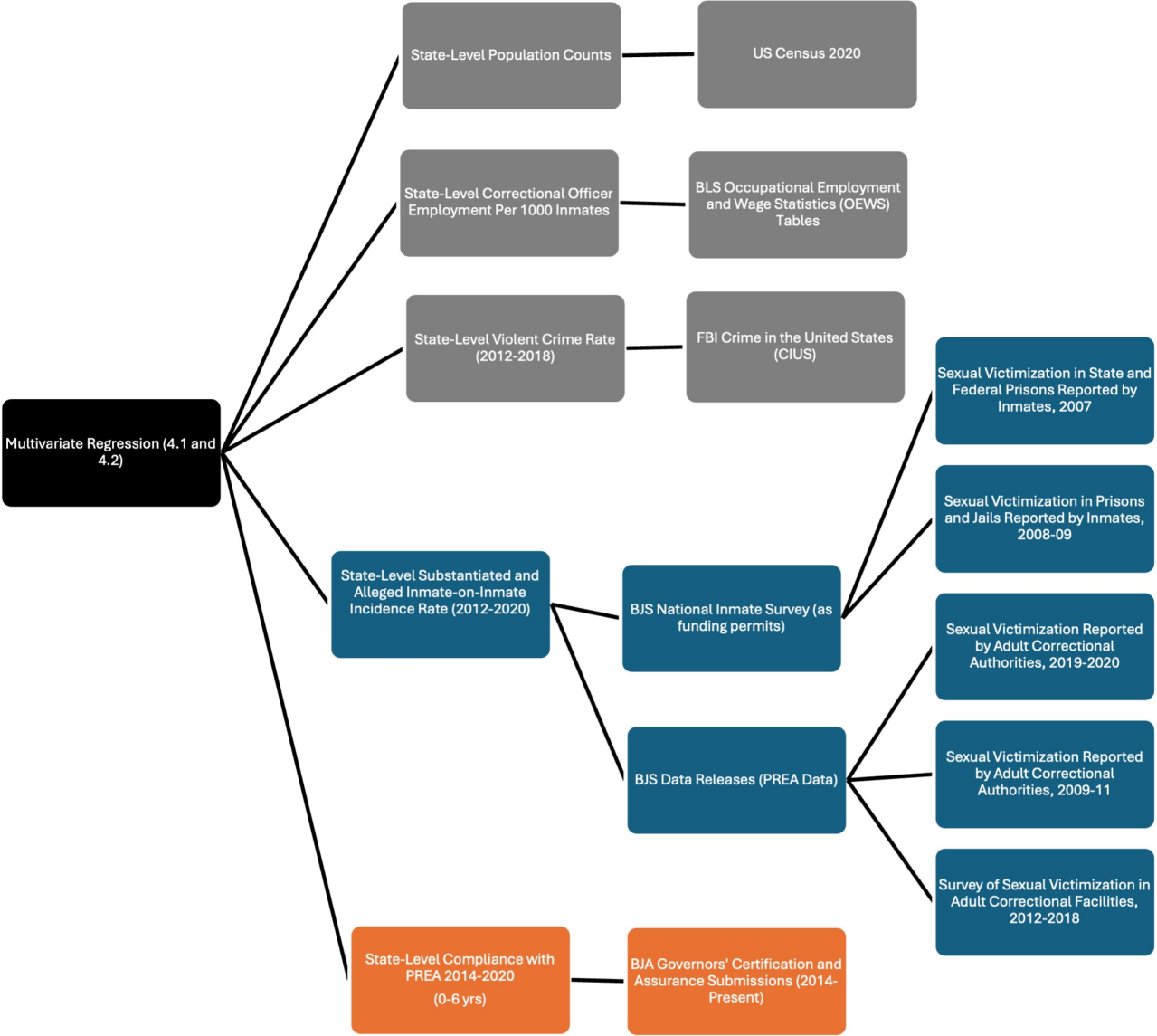


Figure 3: Data sources.